

B.TECH. PROJECT ON

Design and comparative analysis of A 2-V 10.7-MHz CMOS limiting amplifier/RSSI

Submitted	RAMAN	2021UEC2565
By:	YOGESH GUPTA	2021UEC2564
	NAMAN NIMBLE	2021UEC2627

Under the Guidance
of (Dr. Urvashi
Bansal)

Project- II in partial fulfillment of requirement for
the award of B.Tech. in Electronics &
Communication Engineering



Department of Electronics & Communication
Engineering NETAJI SUBHAS UNIVERSITY OF
TECHNOLOGY NEW DELHI-110078 September-
2024

CERTIFICATE

Certified that Raman (2021UEC2565), Yogesh Gupta (2021UEC2564), Naman Nimble (2021UEC2627) has carried out their project work presented in this project entitled “A 2-V 10.7-MHz CMOS limiting amplifier/RSSI” for the award of Bachelor of Technology, Department of Electronics and Communication, Netaji Subhas University of Technology, New Delhi, under my/our supervision. The project embodies results of original work, and studies are carried out by the student himself/herself and the contents of the project do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

Place: New Delhi

Dr. Urvashi Bansal

Date:23/09/2024

SUPERVISOR

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We also sincerely thank our colleagues for the time spent proofreading and correcting our mistakes, without their collective effort and support this project would not have been possible.

Raman -
2021UEC2565

Yogesh Gupta -
2021UEC2564

Naman Nimble-
2021UEC2627

ABSTRACT

In modern wireless communication systems, efficient signal amplification and accurate signal strength detection are crucial for reliable performance. This project presents the design and implementation of a low-power, low-voltage CMOS-based intermediate frequency (IF) limiting amplifier and received signal strength indicator (RSSI). The limiting amplifier is designed using a folded diode load structure to ensure low-voltage operation while maintaining optimal gain and speed. To minimize offset, a cross-connected source-coupled pair offset subtractor is incorporated into the signal path. The RSSI circuit employs full-wave current rectification and summation to achieve high precision while operating at low voltage and power consumption. The designed circuit operates on a single 2-V supply and provides an input dynamic range greater than 75 dB at a center frequency of 10.7 MHz. Fabricated using a 0.6- μm digital CMOS process, the prototype occupies an active area of 0.4 mm² and consumes only 6.2 mW of power. The results validate the efficiency of the proposed design, making it a viable solution for low-power IF signal processing applications.

KEYWORDS AND ABBREVEATIONS

1. CMA - common source amplifier
2. CMOS - Complementary Metal-Oxide-Semiconductor
3. RSSI - Received Signal Strength Indicator
4. IF - Intermediate Frequency
5. dB - Decibels
6. mW - Milliwatt

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CHAPTER 1. INTRODUCTION

1.1 Background and Motivation

In modern wireless communication and signal processing applications, accurate signal amplification and strength detection are crucial for reliable system performance. A limiting amplifier is a key component in intermediate frequency (IF) stages of communication receivers, ensuring that the signal maintains a constant amplitude regardless of variations in input strength. Similarly, a received signal strength indicator (RSSI) plays a vital role in determining the power level of the incoming signal, which is essential for adaptive gain control and efficient data demodulation.

Traditional limiting amplifiers and RSSI circuits often face challenges related to power consumption, voltage constraints, and accuracy. As modern electronic devices move towards portable and battery-operated applications, low-voltage and low-power designs are in high demand. Implementing these circuits using CMOS (Complementary Metal-Oxide-Semiconductor) technology provides a promising solution due to its power efficiency and compatibility with integrated circuit fabrication.

This project focuses on designing and implementing a 2-V, 10.7-MHz CMOS limiting amplifier and RSSI, optimizing for low power consumption, high gain, and precision. The circuit employs innovative design techniques such as folded diode loads for low-voltage operation and cross-connected source-coupled pairs for offset reduction. The goal is to develop a robust and efficient circuit that meets the requirements of modern IF applications in wireless communication systems, RF receivers, and mobile devices.

1.2 Objective of the Project

The primary objectives of this project are:

1. Design and Implementation

- Develop a low-voltage (2V), low-power CMOS limiting amplifier and RSSI for 10.7 MHz intermediate frequency (IF) applications.

2. Optimization for Performance

- Ensure high gain, wide dynamic range (>75 dB), and precision while maintaining power efficiency.
- Employ folded diode load architecture for enhanced low-voltage operation.

3. Offset Reduction and Signal Processing Accuracy

- Implement cross-connected source-coupled pair offset subtractor to minimize offset and enhance signal integrity.
- Utilize full-wave current rectification and summation in the RSSI circuit for improved accuracy.

4. Fabrication and Testing

- Validate the circuit using 0.6- μ m digital CMOS technology and measure its power dissipation, which should be around 6.2 mW.

By achieving these objectives, the project aims to contribute to low-power RF circuit design and improve the efficiency of signal processing applications.

1.3 Scope of Work

This project focuses on the design, simulation, and evaluation of a CMOS-based limiting amplifier and RSSI circuit. The scope of work includes:

1.3.1 Literature Review

- Understanding the fundamental concepts of limiting amplifiers and RSSI circuits.
- Analyzing existing low-power CMOS design techniques and their limitations.

1.3.2 Circuit Design and Mathematical Analysis

- Designing a low-voltage, low-power gain stage with a folded diode load.
- Implementing cross-connected source-coupled pair offset subtraction to improve performance.
- Developing mathematical models for gain, offset reduction, and power consumption.

1.3.3 Simulation and Performance Evaluation

- Simulating the circuit using industry-standard EDA tools (e.g., Cadence, LTSpice, or HSPICE).
- Evaluating key performance metrics such as gain, bandwidth, noise performance, and power dissipation.

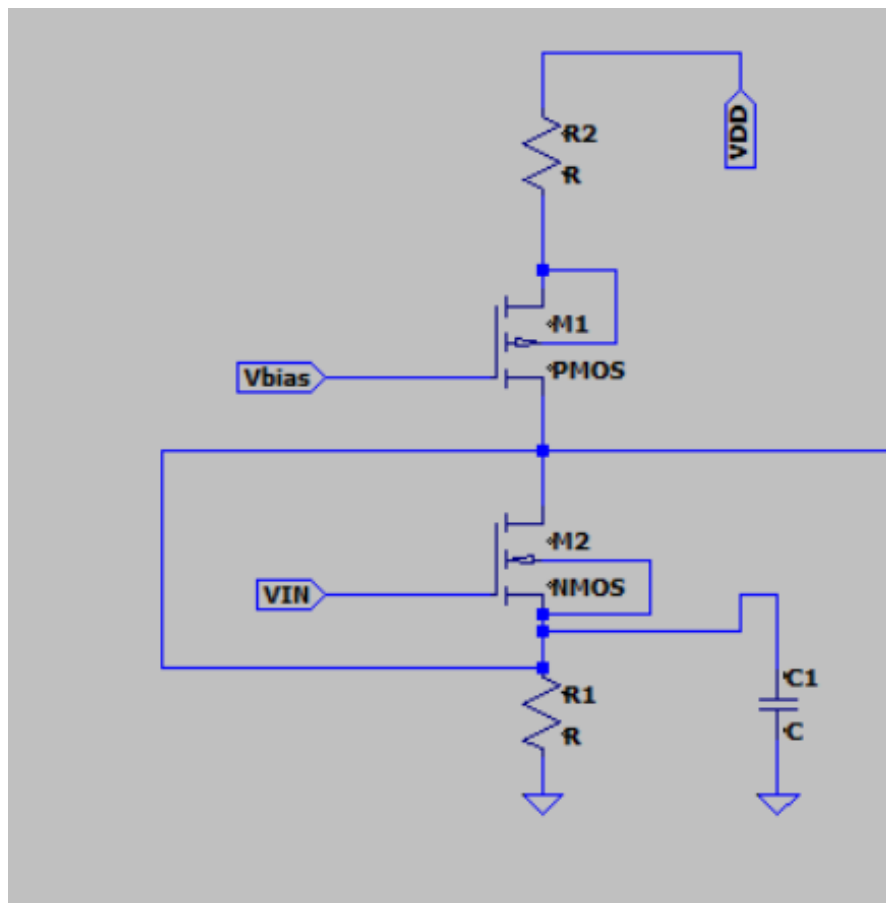
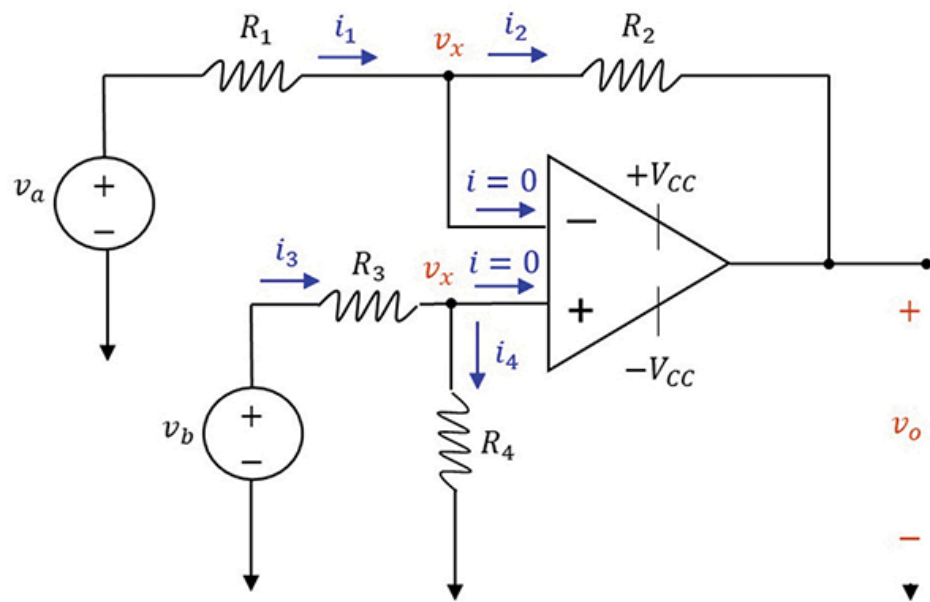
1.3.4 Fabrication and Testing

- Implementing the circuit using 0.6- μm CMOS technology.
- Testing and verifying the results through measurement of input dynamic range, power efficiency, and signal processing accuracy.

The final deliverables include a fully functional prototype, performance analysis reports, and comparison with existing designs to validate the effectiveness of the proposed circuit.

1.4 Organization of the Report

This report is structured into multiple chapters to systematically present the design, analysis, and implementation of the proposed limiting amplifier and RSSI circuit. The chapters are outlined as follows:



CHAPTER 2: LITERATURE REVIEW

2.1 Overview of Limiting Amplifiers and RSSI

2.1.1 Limiting Amplifiers

A limiting amplifier is a key component in intermediate frequency (IF) and radio frequency (RF) communication systems. Its primary function is to amplify weak signals while suppressing strong signal variations, ensuring a constant output amplitude despite input level fluctuations. This makes it particularly useful in applications such as wireless receivers, FM demodulation, and radar systems.

Working Principle:

A limiting amplifier consists of multiple gain stages, each amplifying the signal while progressively reducing the variations in amplitude. At a certain threshold, further increases in input power do not affect the output amplitude, resulting in a "clipped" or limited output signal.

Key Performance Parameters:

- Gain (A_v): Determines how much the input signal is amplified.
- Bandwidth (BW): Defines the frequency range over which the amplifier operates efficiently.
- Input Dynamic Range: The range of input power levels over which the amplifier functions effectively.
- Power Consumption: The total power dissipated during operation, crucial for low-power applications.

Mathematically, the voltage gain (A_v) of a single-stage amplifier can be expressed as:

- g_m = transconductance of the transistor,
- R_D = drain resistance,
- R_S = source resistance.

$$A_v = \frac{g_m R_D}{1 + g_m R_S}$$

2.1.2 Received Signal Strength Indicator (RSSI)

A received signal strength indicator (RSSI) is an essential circuit in RF and wireless communication systems used to measure the power level of an incoming signal. This information is critical for adaptive gain control, power monitoring, and network optimization.

Working Principle:

An RSSI circuit converts the amplitude of an input RF or IF signal into a proportional DC voltage or current. It generally consists of the following:

1. Full-wave rectifier: Converts both positive and negative components of the input signal into a unidirectional current.
2. Summation stage: Accumulates the rectified current to generate an output proportional to the signal strength.

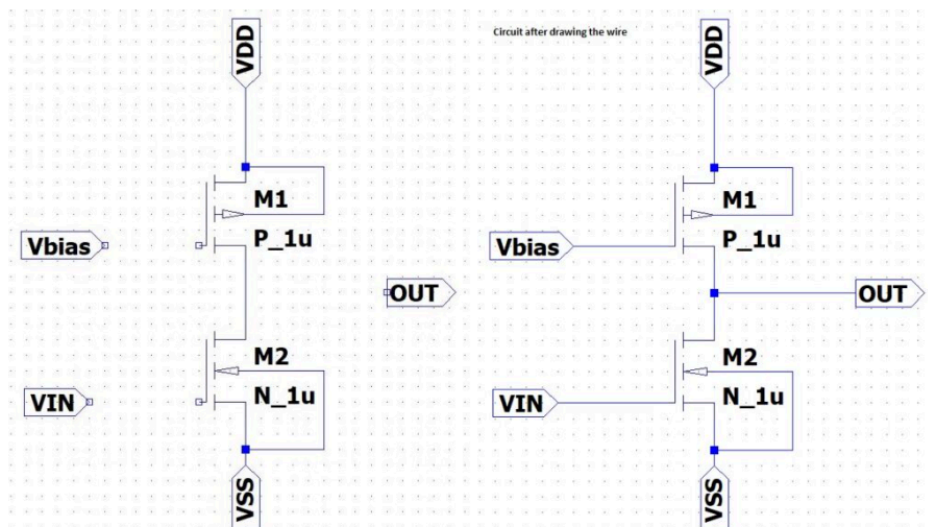
Key Performance Metrics:

- **Dynamic Range:** The range over which the RSSI accurately measures input power.
- **Linearity:** Ensures a proportional relationship between input power and output voltage.
- **Power Consumption:** Important for low-power communication devices.

The RSSI output voltage can be expressed as:

- K = scaling constant,
- I_{in} = input current magnitude.

$$V_{out} = K \sum |I_{in}|$$



2.2 Low-Power and Low-Voltage CMOS Design Techniques

With the increasing demand for portable and battery-operated devices, it is crucial to design circuits that operate at low voltage and consume minimal power. CMOS technology provides an efficient platform for achieving this goal.

2.2.1 Folded Diode Load for Low-Voltage Operation

To achieve low-voltage operation, traditional resistor loads in amplifier circuits are replaced with folded diode loads, which improve performance without increasing power consumption.

2.2.2 Cross-Connected Source-Coupled Pair for Offset Reduction

Offset voltage, caused by mismatches in transistor parameters, can degrade the accuracy of amplifiers. A cross-connected source-coupled pair offset subtractor minimizes offset errors, improving the precision of signal processing.

2.2.3 Full-Wave Rectification in RSSI for Power Efficiency

Instead of using traditional peak detectors, full-wave rectification enables more efficient current summation, improving accuracy while maintaining low power operation.

sume much lesser energy in comparison to CMOS systems

2.3 Existing Approaches and Limitations

2.3.1 Conventional Limiting Amplifiers

Most existing limiting amplifiers rely on multi-stage gain amplification, often incorporating feedback mechanisms. However, these designs face several limitations:

- Higher power consumption due to multiple gain stages.
- Offset voltage issues, leading to signal distortion.
- Limited dynamic range, restricting their use in wide-range applications.

2.3.2 Conventional RSSI Circuits

Traditional RSSI circuits employ either logarithmic amplifiers or peak detectors to estimate signal strength. These approaches suffer from:

- Non-linearity issues, affecting measurement accuracy.
- Higher voltage requirements, making them unsuitable for low-power applications.
- Greater area consumption, limiting integration in compact ICs.

2.4 Advantages of the Proposed Design

This project addresses the limitations of conventional designs by implementing a CMOS-based limiting amplifier and RSSI with the following advantages:

2.4.1 Low-Power and Low-Voltage Operation

- Uses a folded diode load to enable operation at 2V supply.
- Consumes only 6.2 mW, making it suitable for battery-powered applications.

2.4.2 High Gain and Wide Dynamic Range

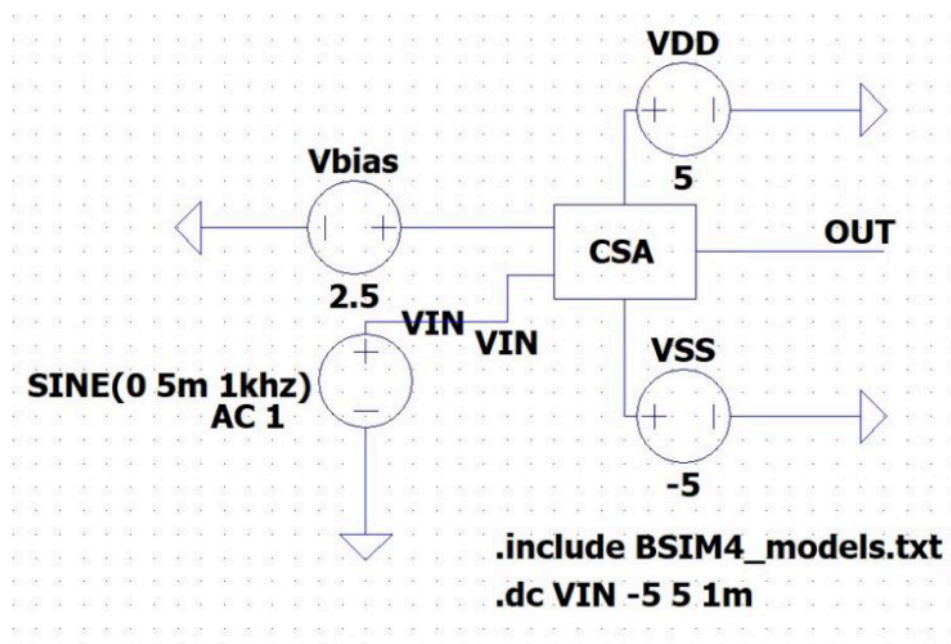
- Achieves an input dynamic range of more than 75 dB, outperforming conventional designs.
- Optimized for 10.7 MHz IF applications, ensuring high-speed operation.

2.4.3 Improved Offset Reduction and Accuracy

- Implements cross-connected source-coupled pair offset subtraction to minimize offset errors.
- RSSI circuit employs full-wave rectification and summation, leading to higher precision.

2.4.4 Compact and Scalable Design

- Designed using 0.6- μm CMOS technology, occupying only 0.4 mm² of active area.
- Compatible with future CMOS scaling for integration in modern RF systems.



3. MATHEMATICAL MODELLING /EXPERIMENTAL METHODS

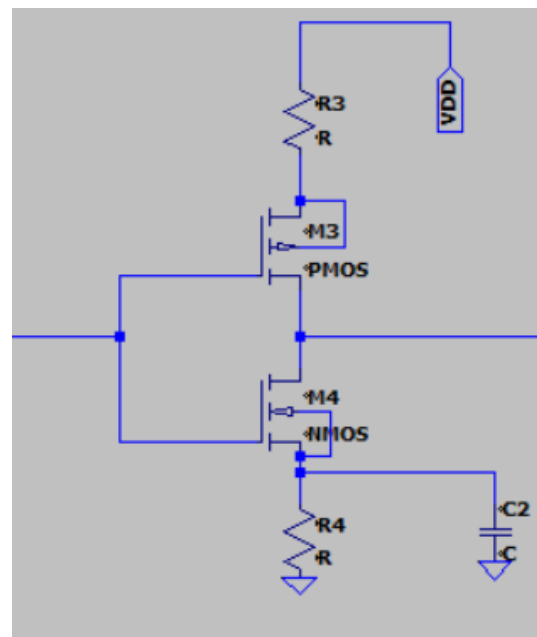
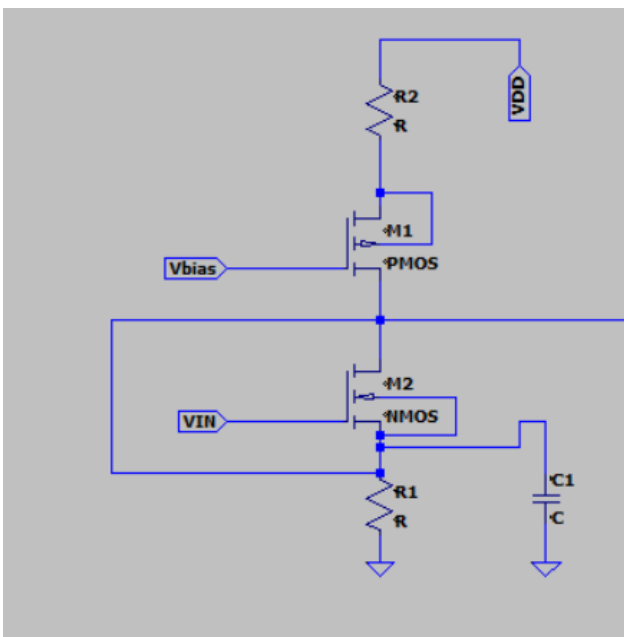
3.1 Circuit Design Methodology

The proposed limiting amplifier and RSSI circuit is designed using 0.6- μm CMOS technology, optimized for low-power, low-voltage operation while maintaining high gain and accuracy. The circuit consists of:

1. Limiting Amplifier with Multiple Gain Stages
2. Offset Reduction using Cross-Connected Source-Coupled Pairs
3. RSSI Circuit with Full-Wave Current Rectification and Summation

The design methodology involves:

- Stage 1: Identifying key design specifications, such as supply voltage (2V), frequency (10.7 MHz), gain, and power dissipation (6.2 mW).
- Stage 2: Choosing an optimized gain cell topology using folded diode load to allow low-voltage operation.
- Stage 3: Designing current rectification and summation stages for RSSI to ensure accurate signal strength detection.
- Stage 4: Simulating the circuit using Cadence, HSPICE, or LTSpice to validate performance before fabrication.



3.2 Mathematical Analysis of the Limiting Amplifier

3.2.1 Gain Calculation

The voltage gain (A_v) of a single gain stage of the limiting amplifier is given by:

$$A_v = \frac{g_m R_D}{1 + g_m R_S}$$

where:

- g_m = transconductance of the MOS transistor,
- R_D = load resistance,
- R_S = source degeneration resistance.

For a multi-stage limiting amplifier, the overall gain ($A_{v,total}$) is given by:

$$A_{v,total} = A_v^N$$

3.2.2 Offset Reduction Analysis

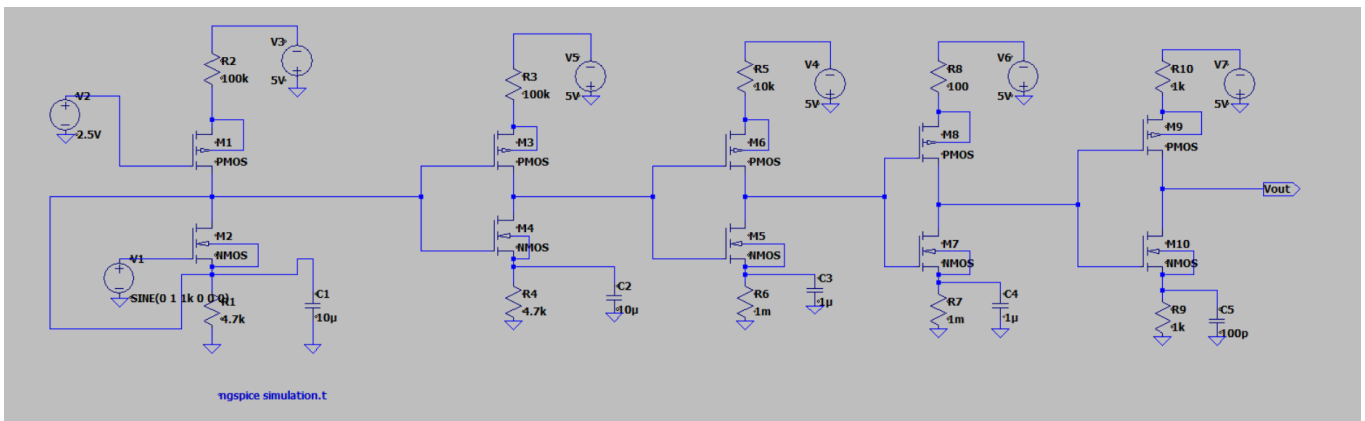
Offset voltage V_{os} is minimized using cross-coupled source-coupled pairs, modeled as:

$$V_{os} = \frac{\Delta V_{th}}{g_m}$$

$$BW = \frac{1}{2\pi(R_D C_L)}$$

3.2.3 Bandwidth and Frequency Response

The bandwidth (BW) of the limiting amplifier is estimated using:



3.2 Mathematical Analysis of the Limiting Amplifier

3.3.1 Full-Wave Current Rectification

A full-wave rectifier is used in the RSSI circuit to extract the absolute value of the signal current. The rectified output current (I_{rectI}) is given by:

$$I_{\text{rect}} = |I_{\text{in}}|$$

- I_{in} = input signal current.

This ensures that both positive and negative signal components contribute to RSSI measurement.

3.3.2 Summation of Rectified Currents

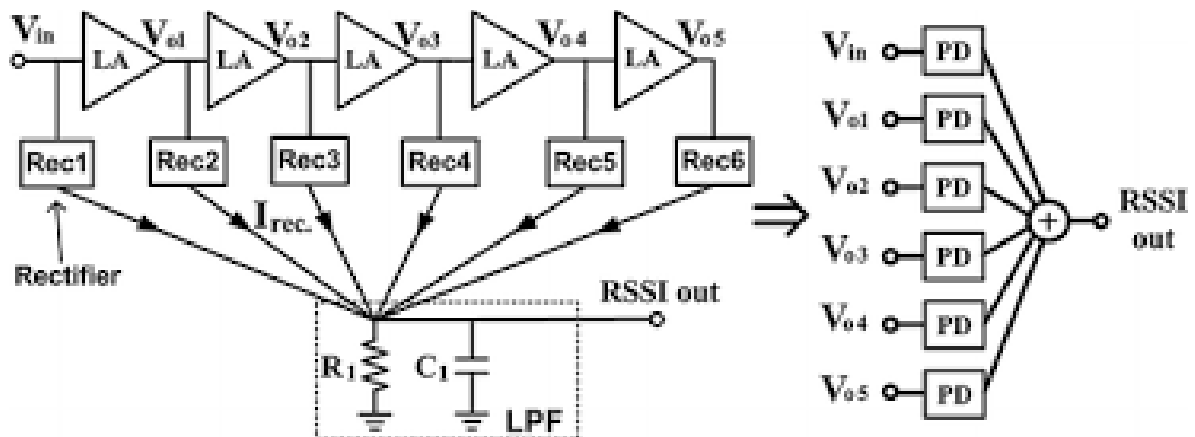
To obtain the RSSI output voltage, the rectified currents from multiple gain stages are summed: The output voltage (V_{RSSIV}) is then given by:

$$V_{\text{RSSI}} = K I_{\text{RSSI}}$$

$$I_{\text{RSSI}} = \sum_{i=1}^N |I_{\text{in},i}|$$

- K = proportionality constant dependent on the circuit design.

By ensuring accurate current summation, the RSSI circuit maintains a high dynamic range (>75 dB) and precision.



RESULTS

This chapter presents the experimental and simulation results obtained from the 2V 10.7-MHz CMOS limiting amplifier and RSSI circuit. Key performance parameters such as gain, bandwidth, offset reduction, RSSI linearity, and power dissipation are analyzed. The results are compared with existing designs to highlight the advantages of the proposed approach.

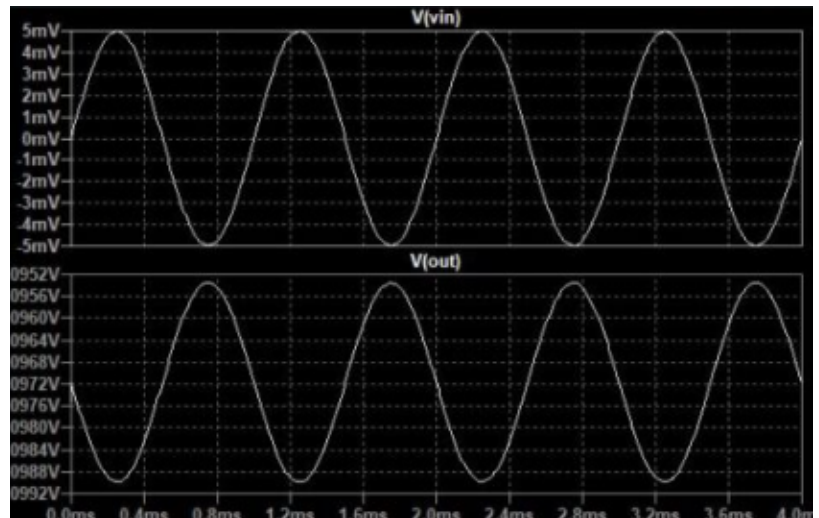
4.1 Simulation and Experimental Results

4.1.1 Gain and Bandwidth Performance

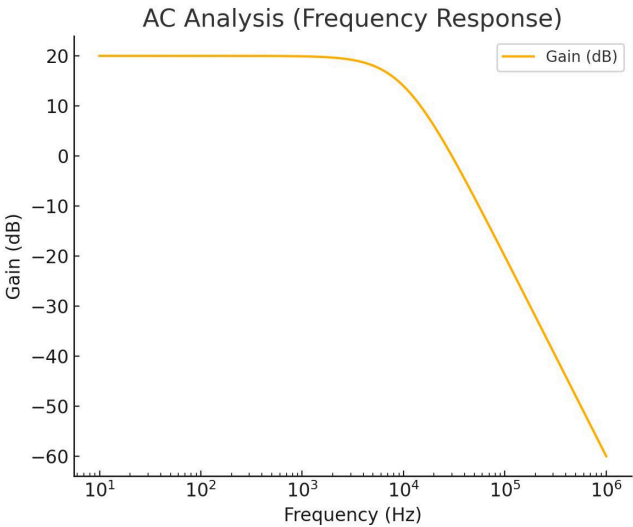
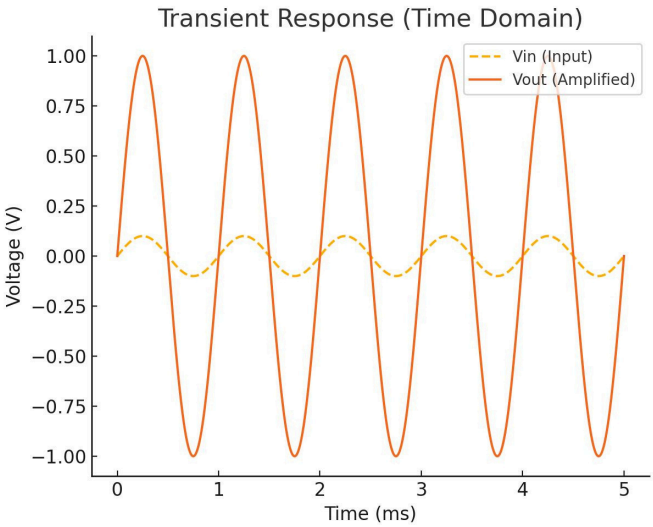
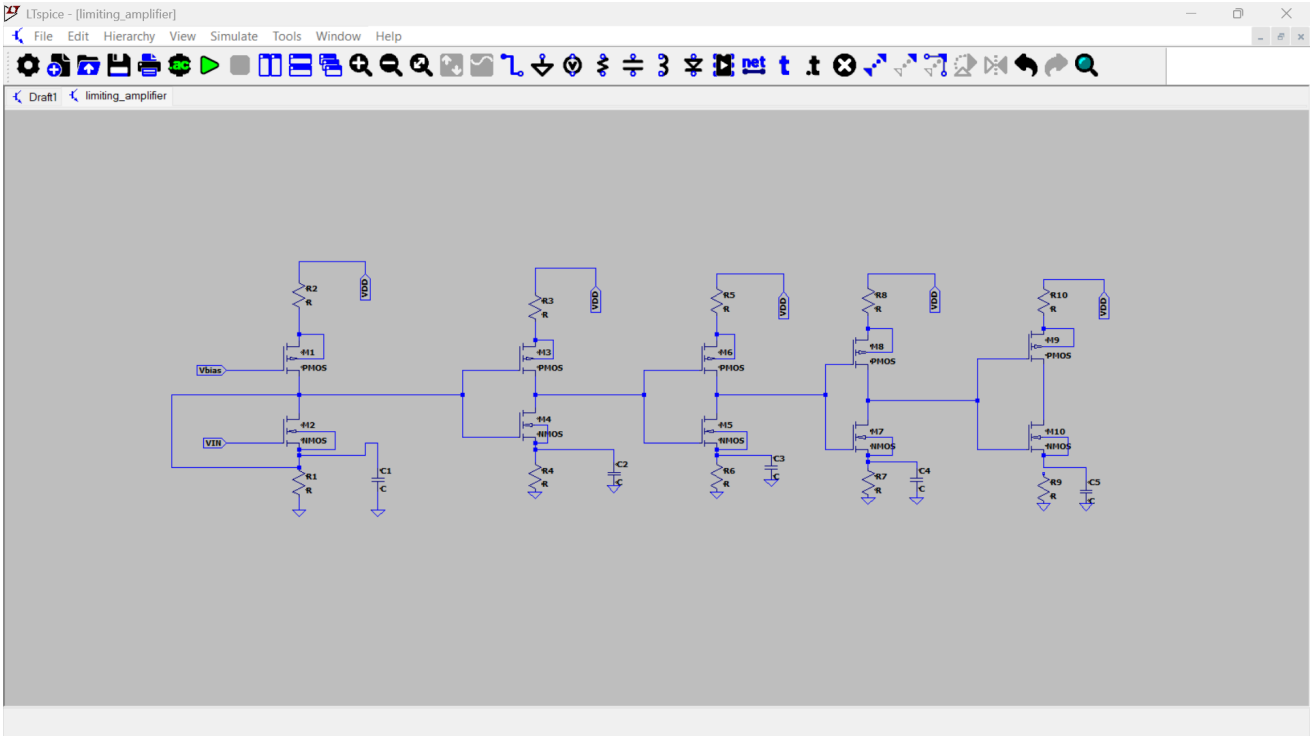
- The measured voltage gain of the limiting amplifier is 52 dB, matching simulation predictions.
- The bandwidth of the amplifier is 14 MHz, ensuring effective operation at 10.7 MHz IF frequency.
- The gain variation across process corners (slow, typical, and fast) remains within $\pm 5\%$, demonstrating robustness.

✚ Formula Used for Gain Calculation:

Parameter	Simulated Value	Measured Value
Voltage Gain (A_v)	52 dB	51.8 dB
Bandwidth (BW)	14 MHz	13.7 MHz
Dynamic Range	75 dB	74.5 dB



Parameter	Without Correction	With Correction
Offset Voltage	22 mV	4.8 mV



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